Unit 5: Semiconductor Devices

Topic 1: Energy Bands and Intrinsic Semiconductors

Definition: Materials are classified into conductors, insulators, and semiconductors based on their energy band gap (the empty energy space electrons must cross to conduct electricity). An intrinsic semiconductor is a pure semiconductor material like pure Silicon or Germanium.

3 Features:

• Conductors have overlapping bands, insulators have a massive band gap, and semiconductors have a very small gap.

• At absolute zero temperature, pure semiconductors hold their electrons tightly and behave like perfect insulators.

• As the temperature increases, their electrical conductivity rises because more electrons gain energy to jump the small gap.

3 Uses:

• Used to manufacture high-precision temperature sensors (thermistors).

• Forms the fundamental base substrate for manufacturing all integrated circuits (ICs).

• Used in specialized physics laboratory apparatus to study basic quantum electron movement.

Real-Life Application: A digital medical thermometer uses the temperature-sensitive behavior of a pure semiconductor element to read body heat accurately.

Topic 2: Extrinsic Semiconductors (P-Type and N-Type)

Definition: An extrinsic semiconductor is a pure semiconductor that has been mixed with a tiny amount of chemical impurity in a process called doping to drastically increase its electrical conductivity.

3 Features:

• N-type semiconductors are doped with pentavalent impurities (like Phosphorus), creating extra free electrons.

• P-type semiconductors are doped with trivalent impurities (like Boron), creating missing electron spaces called holes.

• The overall doped material remains electrically neutral despite having excess free charge carriers.

3 Uses:

• Used to construct transistors which amplify radio and microphone audio signals.

• Forms the essential building blocks for making microprocessors inside computing devices.

• Used in making specialized electronic sensors that detect magnetic fields.

Real-Life Application: The silicon chips running inside your electronic smart watch are made of tightly arranged P-type and N-type semiconductor sections.

Topic 3: P-N Junction Diode and Rectifiers

Definition: A P-N junction diode is a basic electronic device formed by joining a P-type and an N-type semiconductor together, allowing electric current to flow in only one direction.

3 Features:

• Creates a tiny, insulating middle zone at the boundary called a depletion region.

• Forward bias (connecting positive to P-type) shrinks the depletion zone and lets current flow easily.

• Reverse bias blocks current completely unless the voltage exceeds a high breakdown limit.

3 Uses:

• Used as a rectifier to safely convert dangerous Alternating Current (AC) into Direct Current (DC).

• Protects delicate electronic circuits from damage if batteries are accidentally put in backward.

• Used as radio wave detectors to pull sound out of wireless broadcasting signals.

Real-Life Application: Your mobile phone charger contains a bridge rectifier circuit made of four diodes that changes 230V AC home wall power into stable 5V DC power.

Topic 4: Special Diodes (Zener Diode, LED, and Solar Cells)

Definition: These are highly customized semiconductor diodes designed to perform unique operations like regulating voltage, emitting bright light, or absorbing sunlight to generate power.

3 Features:

• A Zener diode operates safely in reverse bias to keep a steady output voltage.

• A Light Emitting Diode (LED) directly releases energy as bright light when current flows forward.

• A Solar Cell absorbs light photons to create free electrons and holes, generating a clean electrical voltage.

3 Uses:

• Zener diodes are used as voltage regulators to protect computers from sudden power surges.

• LEDs are used for highly efficient, long-lasting household bulbs and indicator lights.

• Solar cells are packed together to build eco-friendly solar panels for homes.

Real-Life Application: The glowing red or green standby power light on your television set is a highly efficient LED.

Topic 5: Logic Gates and Digital Electronics

Definition: A logic gate is an electronic switching circuit that makes logical decisions based on the combination of digital input signals (0 for OFF, 1 for ON).

3 Features:

• Basic types include AND (all inputs must be 1), OR (any input can be 1), and NOT (inverts the signal).

• Universal gates like NAND and NOR can be combined to build any other type of gate circuit.

• Their exact mathematical behaviors are tracked using simple standard tables called truth tables.

3 Uses:

• Forms the fundamental logic processing pathways inside computer processors (CPUs).

• Used to design automated security alarm systems and digital calculators.

• Helps program industrial robotic arms to perform specific step-by-step tasks.

Real-Life Application: An automated bank ATM only dispenses cash if you insert a valid card AND enter the correct PIN (an application of an AND logic gate circuit).